Stage-by-Stage Impact of Blind Hiring

1 Summary

Core Thesis:

The traditional justification for blind hiring is built on the assumption of *in-group favoritism* — the premise that majority gatekeepers harbor implicit biases against underrepresented minorities (URMs) and systematically favor candidates who look like themselves. Blind hiring is theoretically deployed to level the playing field by removing demographic noise'' and revealing pure technicalsignal.’’

Our experiment demonstrates the *exact opposite* is occurring in modern software engineering hiring.

Because majority gatekeepers organically apply an informal out-group premium'' to minority candidates, blinding demographic information fundamentally reshuffles hiring outcomes --- mechanically harming URMs and boosting White Males. Furthermore, this informal bias is highly elastic to formal corporate policy. When firms lack explicit diversity mandates, individual evaluators engage in \emph{compensatory bias}, informally boosting minority scores. Enforcing blind hiring in these environments destroys this potential safety net. Finally, despite claims that demographics are distractingnoise,’’ blinding resumes yielded *no improvement* in the actual technical quality of the selected candidates.

2 Overall Treatment Effect on Evaluations (Reshuffling)

- **The Finding:** Blinding demographic information during candidate evaluation *reshuffles* hiring outcomes by disrupting a pre-existing system of *demographic premiums and penalties.*
- **The Winners: White Males.** Evaluated by Engineers, White Male overall scores mechanically rise by +8.5% of a SD under blinding.
- **The Losers: Hispanic candidates.** When stripped of their demographic identity, Hispanic Male scores plunge by -11.5% of a SD (evaluated by HR), and Hispanic Female scores fall by -6.9% of a SD (evaluated by Engineers).
- **The Takeaway:** Blind hiring removes a *baseline penalty* that White Males were experiencing in the non-blind evaluations, while simultaneously stripping Hispanic candidates of an *(diversity) premium.*
- **Caveat:** Hiring is often an *ordinal ranking* problem — one candidate's gain corresponds to another's lost opportunity — so the absence of statistical significance on a given row does not mean the corresponding group is unaffected by blinding.

Treatment Effects of Blinding for	Panel A: HR	Panel B: Engineer
	HR	Engineer
White Male	0.037	0.085***
	(0.057)	(0.028)
White Female	0.013	0.023
	(0.051)	(0.036)
Asian Male	0.015	−0.018
	(0.059)	(0.028)
Asian Female	−0.023	−0.029
	(0.048)	(0.029)
Hispanic Male	−0.115**	−0.003
	(0.048)	(0.030)
Hispanic Female	0.087*	−0.069**
	(0.052)	(0.034)
Resume (Content) FE	✓	✓
Display Position FE	✓	✓
Respondent FE	×	×
Respondent Characteristics	✓	✓
Observations	2466	2772
R^2	0.423	0.743
Adj. R^2	0.408	0.737

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Standard errors clustered by respondent.

3 Rejecting Traditional In-Group Favoritism (Why White Male Wins Under Blinding)

- **The Context:** White Males constitute the overwhelming majority of Software Engineer evaluators in this sample (and the industry at large). *Legacy theories assume this group favors its own.*
- **The Finding:** Our experiment shows that the modern tech industry actually operates on a preference for the *out-group* (or preference for *under-represented groups*). Because White Males make up the majority of the evaluator pool, their behavior drives the aggregate metrics. When you blind resumes, the scores of concordant (race/gender) applicants evaluated by Engineers actually rise by +6.9% of a SD ($p < 0.05$) while the scores of discordant applicants fall by -0.9% of a SD ($p < 0.05$).
- **The Takeaway:** This shows that, under non-blinded evaluations, the majority gatekeepers were *actively penalizing* candidates who looked like themselves and assigning a *premium to evaluation scores of minority candidates.* Enforcing blind hiring doesn't stop discrimination against minorities; it stops the White Male majority from systematically boosting them. This explains why, in our experiment, White Male applicants are the primary beneficiaries of blind hiring: they are "rescued" from the baseline penalty imposed by their own demographic engineer peers.

	Concordant Race		Concordant Gender		Concordant Race AND Gender	
	HR	Eng	HR	Eng	HR	Eng
Concordant Applicant	−0.020	0.045*	−0.010	0.029*	−0.076	0.069**
	(0.042)	(0.023)	(0.027)	(0.015)	(0.068)	(0.032)
Nonconcordant Applicant	0.007	−0.014*	0.009	−0.030*	0.011	−0.009**
	(0.015)	(0.007)	(0.025)	(0.016)	(0.010)	(0.004)
Resume (Content) FE	✓	✓	✓	✓	✓	✓
Display Position FE	✓	✓	✓	✓	✓	✓
Respondent FE	×	×	×	×	×	×
Respondent Characteristics	✓	✓	✓	✓	✓	✓
Observations	2466	2772	2466	2772	2466	2772
R^2	0.421	0.742	0.421	0.742	0.421	0.742
Adj. R^2	0.407	0.737	0.407	0.737	0.408	0.737

* p < 0.1, ** p < 0.05, *** p < 0.01

Standard errors clustered by respondent.

4 Institutional Substitution (Firm Policy vs. Individual Bias)

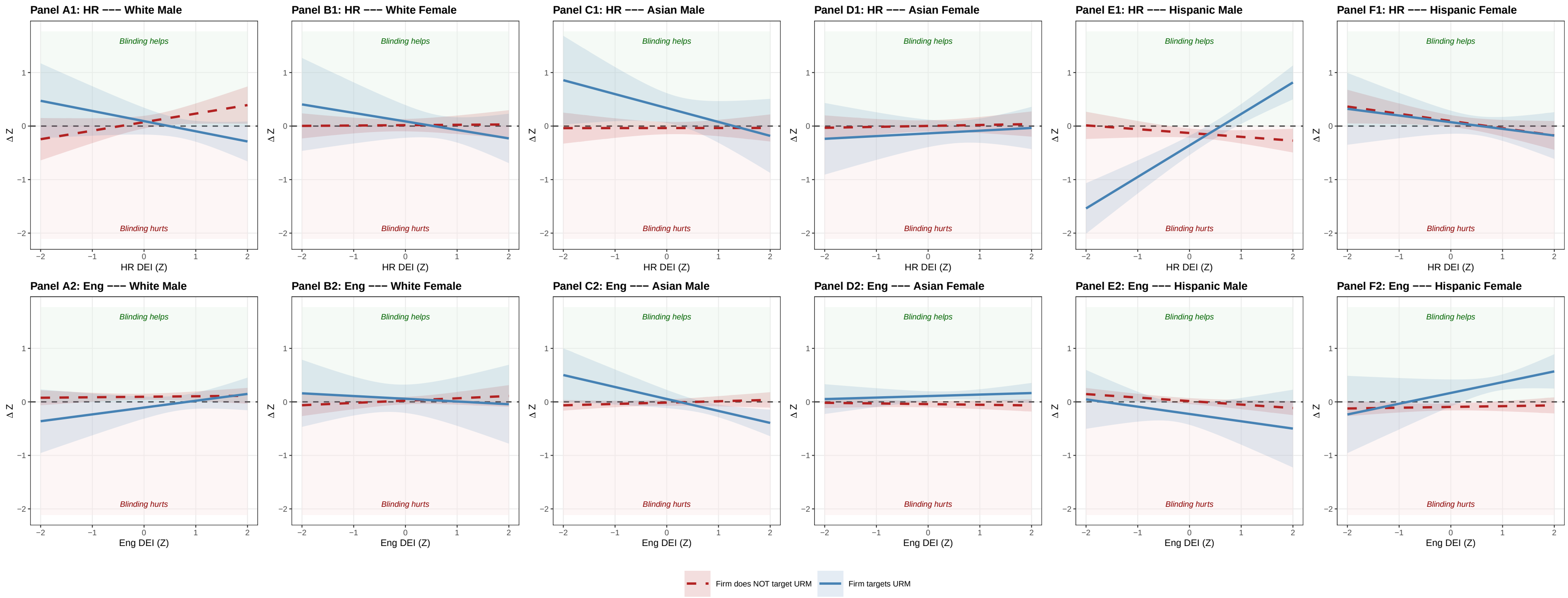
- **The Puzzle of Individual Ideology (Section 4.1):** If one merely examines the two-way relationship between an evaluator's personal DEI priority and their evaluation scores, the effects appear surprisingly *muted and statistically noisy*. At first glance, one might incorrectly conclude that personal ideology has little impact on how blind hiring reshuffles outcomes.
  - **Compensatory Bias** (Firms with NO Diversity Mandate, see red lines): When a firm lacks an explicit objective policy targeting URMs, highly pro-DEI HR professionals engage in *compensatory bias*. They fill the perceived institutional void by informally inflating minority candidates' scores in the baseline.
  - **Institutional Crowding-Out** (Firms WITH a Diversity Mandate, see blue lines): Conversely, when the firm explicitly mandates URM targeting, this formal policy *crowds out* the evaluator's informal efforts. The pro-DEI HR professional, when combined with institutional DEI targeting policy, *mutes* from assigning premium to under-represented minority's scores.
- (Note *Divergence between HR and Engineer Results*, particularly in Triple Interaction.)

4.1 (Combined) Per-Race-Gender Heterogeneity by DEI Preference, Firm URM Targeting, and Firm DEI Attention

- **Individual DEI Index (Z)** — equal-weight average (1/5 each) of five z-scored items: the four stated DEI views plus the revealed-priority ranking of "Diversity and Representation" from the seven-goals question.
- **Firm Targets Fem/His (0/1)** — indicator for whether the respondent's firm has explicit policies targeting Female or Hispanic candidates.
- **Firm DEI Attn Index (Z)** — within-cell z-score of the equal-weight average of the respondent's perceived firm attention to race and gender DEI.

	Simple DEI Heterogeneity		Triple: Firm Targets URM		Triple: Firm Pays DEI Attention	
	HR	Engineer	HR	Engineer	HR	Engineer
White Male	0.040 (0.056)	0.086*** (0.028)	0.074 (0.060)	0.096*** (0.029)	0.095 (0.058)	0.056* (0.029)
White Female	0.014 (0.051)	0.023 (0.035)	0.017 (0.059)	0.024 (0.036)	−0.002 (0.070)	0.028 (0.035)
Asian Male	0.016 (0.060)	−0.017 (0.028)	−0.037 (0.059)	−0.014 (0.030)	0.080 (0.069)	−0.025 (0.034)
Asian Female	−0.023 (0.048)	−0.030 (0.029)	0.003 (0.054)	−0.042 (0.031)	−0.026 (0.063)	−0.018 (0.041)
Hispanic Male	−0.113** (0.048)	−0.004 (0.029)	−0.129*** (0.044)	0.015 (0.029)	−0.171*** (0.057)	0.028 (0.028)
Hispanic Female	0.091* (0.050)	−0.070** (0.033)	0.095* (0.056)	−0.095*** (0.033)	0.062 (0.056)	−0.082* (0.046)
White Male × Individual DEI Index (Z)	0.098 (0.079)	0.009 (0.032)	0.159* (0.090)	0.009 (0.034)	0.165** (0.066)	0.036 (0.043)
White Female × Individual DEI Index (Z)	−0.014 (0.051)	0.039 (0.047)	0.007 (0.057)	0.043 (0.049)	0.002 (0.069)	−0.007 (0.046)
Asian Male × Individual DEI Index (Z)	−0.007 (0.064)	0.015 (0.026)	0.001 (0.063)	0.025 (0.028)	−0.026 (0.066)	0.046 (0.047)
Asian Female × Individual DEI Index (Z)	0.009 (0.047)	−0.005 (0.022)	0.017 (0.053)	−0.012 (0.022)	−0.005 (0.061)	0.009 (0.049)
Hispanic Male × Individual DEI Index (Z)	0.020 (0.063)	−0.077*** (0.029)	−0.071 (0.057)	−0.065** (0.028)	−0.033 (0.068)	−0.096** (0.041)
Hispanic Female × Individual DEI Index (Z)	−0.136** (0.062)	0.038 (0.034)	−0.135* (0.069)	0.015 (0.033)	−0.153*** (0.058)	0.009 (0.065)
White Male × Firm Targets Fem/His (0/1)			0.019 (0.143)	−0.202* (0.109)		
White Female × Firm Targets Fem/His (0/1)			0.070 (0.167)	0.034 (0.139)		
Asian Male × Firm Targets Fem/His (0/1)			0.374** (0.153)	0.068 (0.091)		
Asian Female × Firm Targets Fem/His (0/1)			−0.139 (0.143)	0.150*** (0.055)		
Hispanic Male × Firm Targets Fem/His (0/1)			−0.232** (0.103)	−0.241** (0.107)		
Hispanic Female × Firm Targets Fem/His (0/1)			−0.023 (0.125)	0.262* (0.133)		
White Male × Firm DEI Attn Index (Z)					0.001 (0.100)	0.013 (0.028)
White Female × Firm DEI Attn Index (Z)					−0.094 (0.065)	0.032 (0.044)
Asian Male × Firm DEI Attn Index (Z)					0.141** (0.068)	−0.048 (0.042)
Asian Female × Firm DEI Attn Index (Z)					−0.018 (0.072)	−0.044 (0.038)
Hispanic Male × Firm DEI Attn Index (Z)					0.116* (0.065)	0.020 (0.027)
Hispanic Female × Firm DEI Attn Index (Z)					−0.125* (0.073)	0.036 (0.054)
White Male × Individual DEI Index (Z) × Firm Targets Fem/His (0/1)			−0.349** (0.156)	0.118 (0.113)		
White Female × Individual DEI Index (Z) × Firm Targets Fem/His (0/1)			−0.166 (0.168)	−0.094 (0.168)		
Asian Male × Individual DEI Index (Z) × Firm Targets Fem/His (0/1)			−0.261 (0.193)	−0.249*** (0.094)		
Asian Female × Individual DEI Index (Z) × Firm Targets Fem/His (0/1)			0.033 (0.135)	0.041 (0.060)		
Hispanic Male × Individual DEI Index (Z) × Firm Targets Fem/His (0/1)			0.660*** (0.108)	−0.071 (0.158)		
Hispanic Female × Individual DEI Index (Z) × Firm Targets Fem/His (0/1)			0.010 (0.151)	0.187 (0.131)		
White Male × Individual DEI Index (Z) × Firm DEI Attn Index (Z)					−0.184* (0.100)	0.024 (0.030)
White Female × Individual DEI Index (Z) × Firm DEI Attn Index (Z)					0.010 (0.073)	0.011 (0.043)
Asian Male × Individual DEI Index (Z) × Firm DEI Attn Index (Z)					−0.185** (0.072)	0.003 (0.029)
Asian Female × Individual DEI Index (Z) × Firm DEI Attn Index (Z)					0.029 (0.066)	−0.024 (0.030)
Hispanic Male × Individual DEI Index (Z) × Firm DEI Attn Index (Z)					0.138* (0.070)	−0.049** (0.023)
Hispanic Female × Individual DEI Index (Z) × Firm DEI Attn Index (Z)					0.181** (0.087)	0.041 (0.049)
Resume (Content) FE	✓	✓	✓	✓	✓	✓
Display Position FE	✓	✓	✓	✓	✓	✓
Respondent FE	×	×	×	×	×	×
Respondent Characteristics	✓	✓	✓	✓	✓	✓
Observations	2466	2772	2466	2772	2160	2358

Standard errors clustered by respondent.



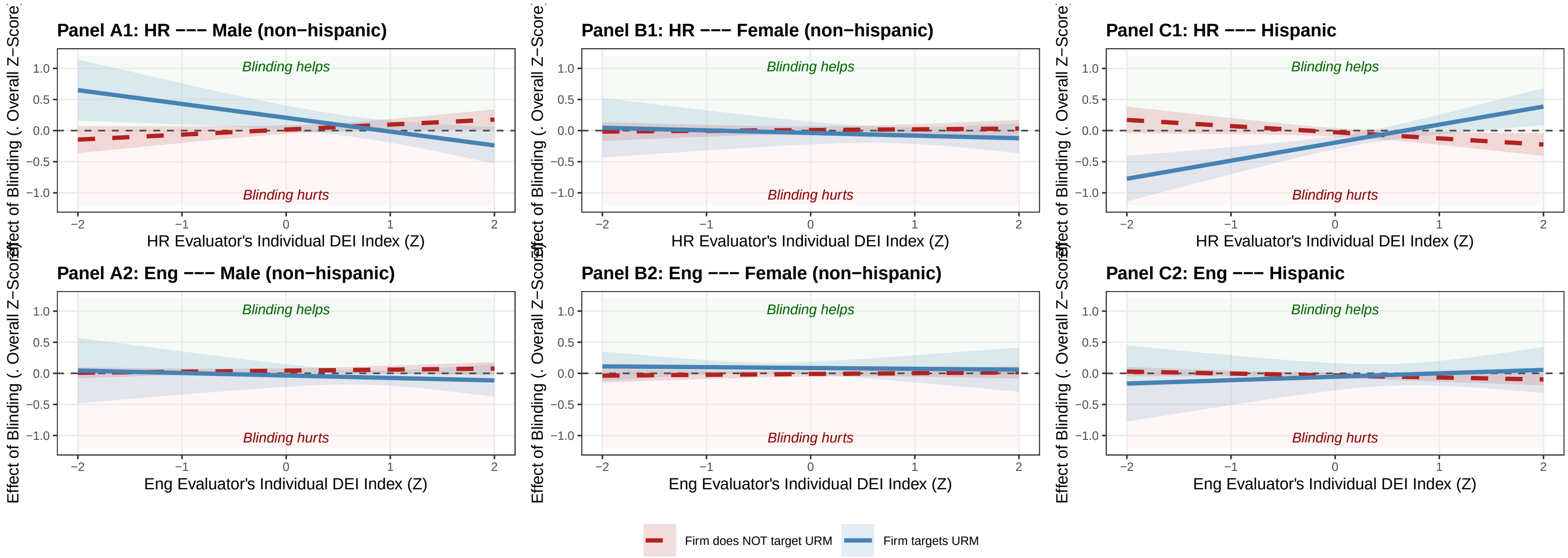
Marginal-effects "scissors" plot built from the §4.4 columns 3-4 (HR & Engineer triple with Firm Targets Fem/His), per race-gender. Top row: HR evaluators (Panels A1–F1). Bottom row: Engineer evaluators (Panels A2–F2). Lines: predicted effect of blinding on each group's overall score as a function of the rater's Individual DEI Index (Z); shaded bands are 95% CIs (SE clustered by respondent).

4.2 (Combined) URM-Axis Heterogeneity by DEI Preference, Firm URM Targeting, and Firm DEI Attention

	Simple DEI Heterogeneity		Triple: Firm Targets URM		Triple: Firm Pays DEI Attention	
	HR	Engineer	HR	Engineer	HR	Engineer
Male (non-hispanic)	0.026	0.036*	0.016	0.043**	0.085**	0.016
	(0.037)	(0.019)	(0.035)	(0.019)	(0.039)	(0.019)
Female (non-hispanic)	−0.006	−0.005	0.008	−0.010	−0.016	0.004
	(0.028)	(0.020)	(0.031)	(0.021)	(0.036)	(0.024)
Hispanic	−0.020	−0.033	−0.026	−0.035	−0.069**	−0.022
	(0.035)	(0.021)	(0.034)	(0.021)	(0.034)	(0.026)
Male (NH) × Individual DEI Index (Z)	0.048	0.011	0.080*	0.017	0.072*	0.041
	(0.044)	(0.022)	(0.046)	(0.023)	(0.036)	(0.031)
Female (NH) × Individual DEI Index (Z)	−0.001	0.015	0.012	0.013	0.001	0.002
	(0.031)	(0.024)	(0.034)	(0.025)	(0.037)	(0.033)
Hispanic × Individual DEI Index (Z)	−0.048	−0.028	−0.099**	−0.032	−0.079*	−0.047
	(0.046)	(0.019)	(0.048)	(0.020)	(0.043)	(0.037)
Male (NH) × Firm Targets Fem/His (0/1)			0.190*	−0.077		
			(0.107)	(0.095)		
Female (NH) × Firm Targets Fem/His (0/1)			−0.047	0.096*		
			(0.100)	(0.054)		
Hispanic × Firm Targets Fem/His (0/1)			−0.168***	−0.020		
			(0.064)	(0.113)		
Male (NH) × Firm DEI Attn Index (Z)					0.069	−0.017
					(0.052)	(0.024)
Female (NH) × Firm DEI Attn Index (Z)					−0.057	−0.007
					(0.038)	(0.026)
Hispanic × Firm DEI Attn Index (Z)					−0.011	0.027
					(0.048)	(0.027)
Male (NH) × Individual DEI Index (Z) × Firm Targets Fem/His (0/1)			−0.303***	−0.056		
			(0.101)	(0.097)		
Female (NH) × Individual DEI Index (Z) × Firm Targets Fem/His (0/1)			−0.054	−0.026		
			(0.090)	(0.077)		
Hispanic × Individual DEI Index (Z) × Firm Targets Fem/His (0/1)			0.389***	0.087		
			(0.095)	(0.118)		
Male (NH) × Individual DEI Index (Z) × Firm DEI Attn Index (Z)					−0.185***	0.013
					(0.057)	(0.022)
Female (NH) × Individual DEI Index (Z) × Firm DEI Attn Index (Z)					0.021	−0.008
					(0.045)	(0.024)
Hispanic × Individual DEI Index (Z) × Firm DEI Attn Index (Z)					0.164***	−0.005
					(0.060)	(0.024)
Resume (Content) FE	✓	✓	✓	✓	✓	✓
Display Position FE	✓	✓	✓	✓	✓	✓
Respondent FE	×	×	×	×	×	×
Respondent Characteristics	✓	✓	✓	✓	✓	✓
Observations	2466	2772	2466	2772	2160	2358

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Standard errors clustered by respondent.



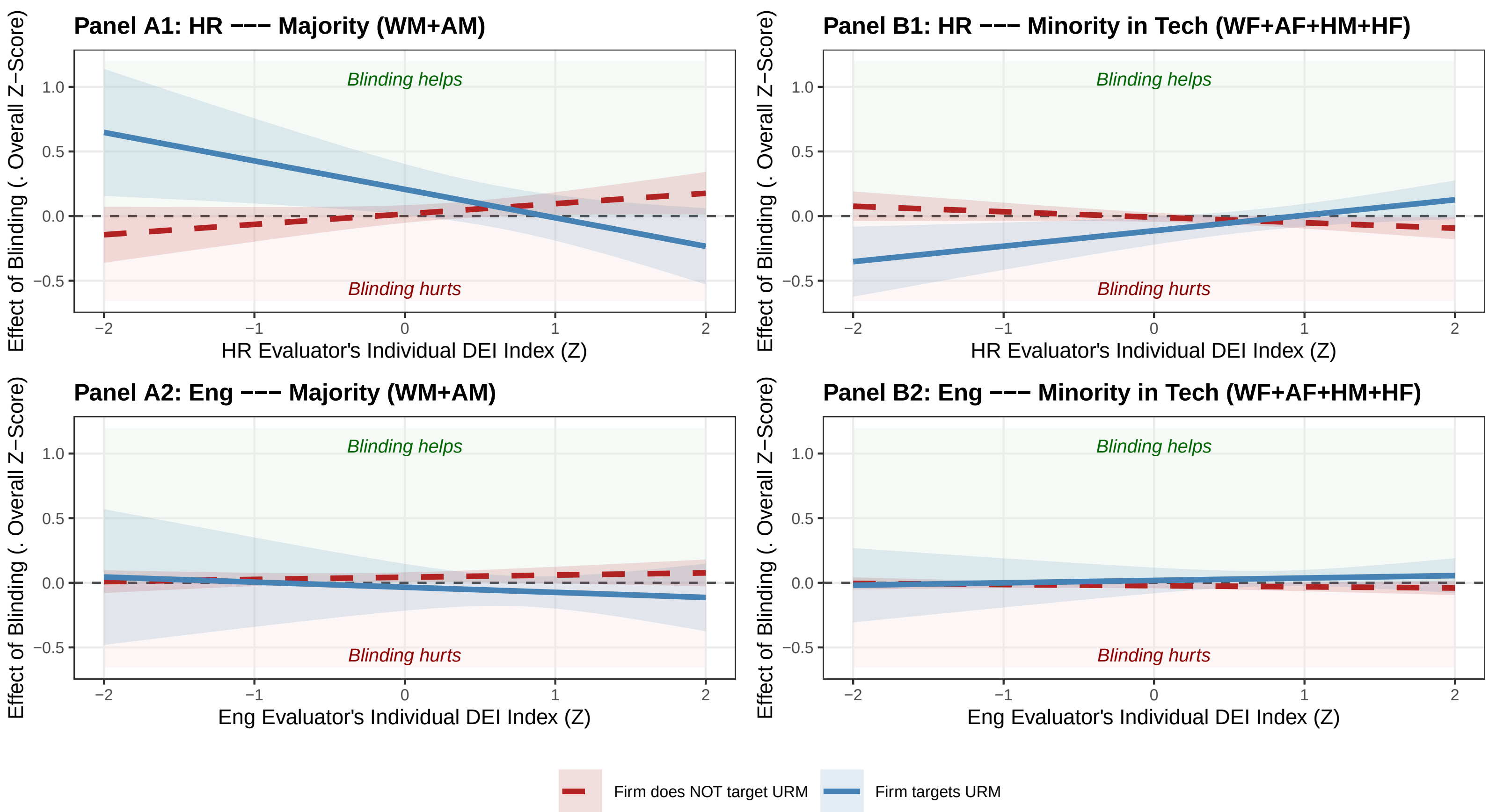
Marginal-effects "scissors" plot built from the §4.4 columns 3-4 (HR & Engineer triple with Firm Targets Fem/His). Top row: HR evaluators (Panels A1, B1, C1). Bottom row: Engineer evaluators (Panels A2, B2, C2). Lines: predicted effect of blinding on each group's overall score as a function of the rater's Individual DEI Index (Z); shaded bands are 95% CIs from the linear combination of axis-triple coefficients (SE clustered by respondent).

4.3 (Combined) URM-Tech Dichotomy by DEI Preference, Firm URM Targeting, and Firm DEI Attention

	Simple DEI Heterogeneity		Triple: Firm Targets URM		Triple: Firm Pays DEI Attention	
	HR	Engineer	HR	Engineer	HR	Engineer
Majority (WM+AM)	0.027	0.036*	0.016	0.043**	0.085**	0.016
	(0.037)	(0.019)	(0.035)	(0.019)	(0.039)	(0.019)
Minority in Tech (WF+AF+HM+HF)	−0.013	−0.019*	−0.009	−0.022**	−0.043**	−0.009
	(0.019)	(0.010)	(0.018)	(0.010)	(0.020)	(0.010)
Majority (WM+AM) × Individual DEI Index (Z)	0.048	0.011	0.080*	0.017	0.071*	0.041
	(0.044)	(0.022)	(0.046)	(0.023)	(0.036)	(0.031)
Minority in Tech (WF+AF+HM+HF) × Individual DEI Index (Z)	−0.025	−0.006	−0.042*	−0.009	−0.038*	−0.022
	(0.023)	(0.012)	(0.024)	(0.012)	(0.019)	(0.016)
Majority (WM+AM) × Firm Targets Fem/His (0/1)			0.191*	−0.077		
			(0.107)	(0.094)		
Minority in Tech (WF+AF+HM+HF) × Firm Targets Fem/His (0/1)			−0.104*	0.041		
			(0.058)	(0.052)		
Majority (WM+AM) × Firm DEI Attn Index (Z)					0.069	−0.017
					(0.052)	(0.024)
Minority in Tech (WF+AF+HM+HF) × Firm DEI Attn Index (Z)					−0.034	0.009
					(0.026)	(0.013)
Majority (WM+AM) × Individual DEI Index (Z) × Firm Targets Fem/His (0/1)			−0.300***	−0.056		
			(0.101)	(0.097)		
Minority in Tech (WF+AF+HM+HF) × Individual DEI Index (Z) × Firm Targets Fem/His (0/1)			0.162***	0.028		
			(0.054)	(0.052)		
Majority (WM+AM) × Individual DEI Index (Z) × Firm DEI Attn Index (Z)					−0.185***	0.013
					(0.057)	(0.022)
Minority in Tech (WF+AF+HM+HF) × Individual DEI Index (Z) × Firm DEI Attn Index (Z)					0.093***	−0.007
					(0.029)	(0.012)
Resume (Content) FE	✓	✓	✓	✓	✓	✓
Display Position FE	✓	✓	✓	✓	✓	✓
Respondent FE	×	×	×	×	×	×
Respondent Characteristics	✓	✓	✓	✓	✓	✓
Observations	2466	2772	2466	2772	2160	2358

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Standard errors clustered by respondent.



Marginal-effects "scissors" plot built from the §4.5 columns 3-4 (HR & Engineer triple with Firm Targets Fem/His) under the URM-Tech dichotomy. Top row: HR evaluators (Panels A1, B1). Bottom row: Engineer evaluators (Panels A2, B2). Lines: predicted effect of blinding on each group's overall score as a function of the rater's Individual DEI Index (Z); shaded bands are 95% CIs (SE clustered by respondent).

5 Treatment Effect on Productivity of Hired (The Efficiency Myth)

Signal Extraction Remains Flat (S5.1.): Removing demographic information (or noise) does *not* meaningfully improve their ability to identify top ability candidates.

Quality of Hires Does Not Improve (S5.2.): Blind hiring has a *null* effect on the actual coding quality of the hired cohort.

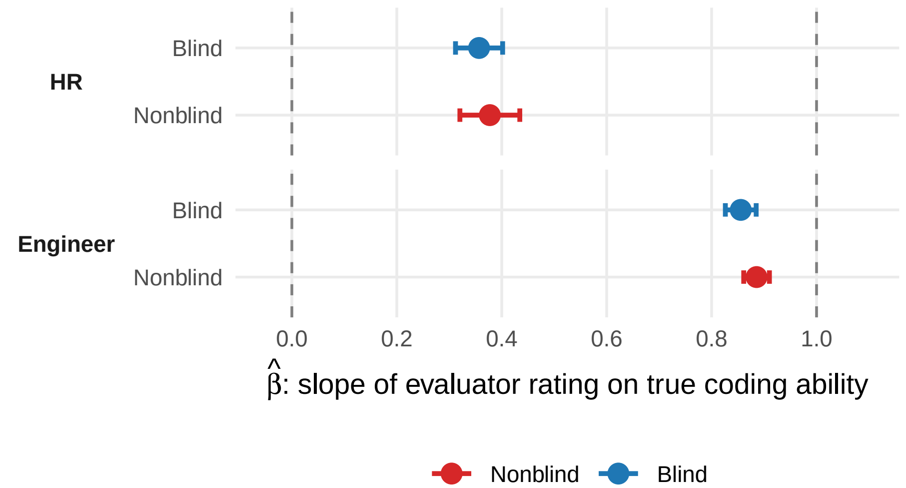
5.1 Does Blinding Lead to More Accurate Coding Skill Evaluations?

Specification:

$$\text{Hiring Evaluation Score (Z)} \sim \text{True Coding Ability (Z)} \times \mathbb{1}[\text{Blind arm}] \mid \text{Respondent FE} + \text{Display Position FE}.$$

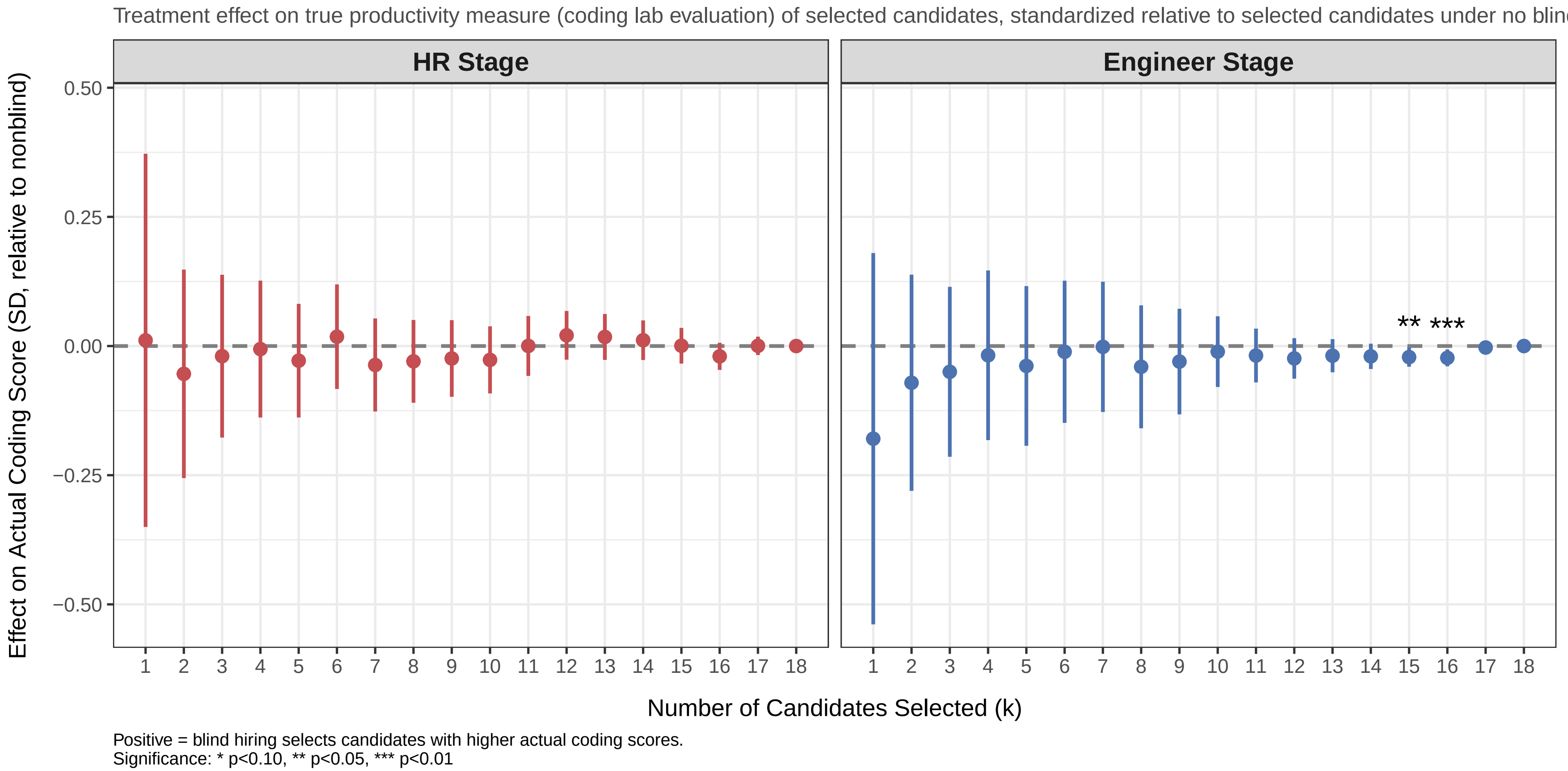
- $H_0 : \beta = 0$ — the evaluator extracts *no signal* from true coding ability.
- $H_0 : \beta = 1$ — the evaluator is *perfectly calibrated*: a 1 SD rise in true ability produces exactly a 1 SD rise in the rating, with no shrinkage.

	HR	Engineer
$\hat{\beta}_{\text{nonblind}}$ (slope, nonblind arm)	0.377 (0.029)	0.886 (0.012)
$\hat{\beta}_{\text{blind}}$ (slope, blind arm)	0.357 (0.023)	0.856 (0.015)
Observations	2,466	2,772
R^2	0.142	0.759
Adj. R^2	0.085	0.743
Resume (Content) FE	×	×
Display Position FE	✓	✓
Respondent FE	✓	✓
Respondent Characteristics	×	×



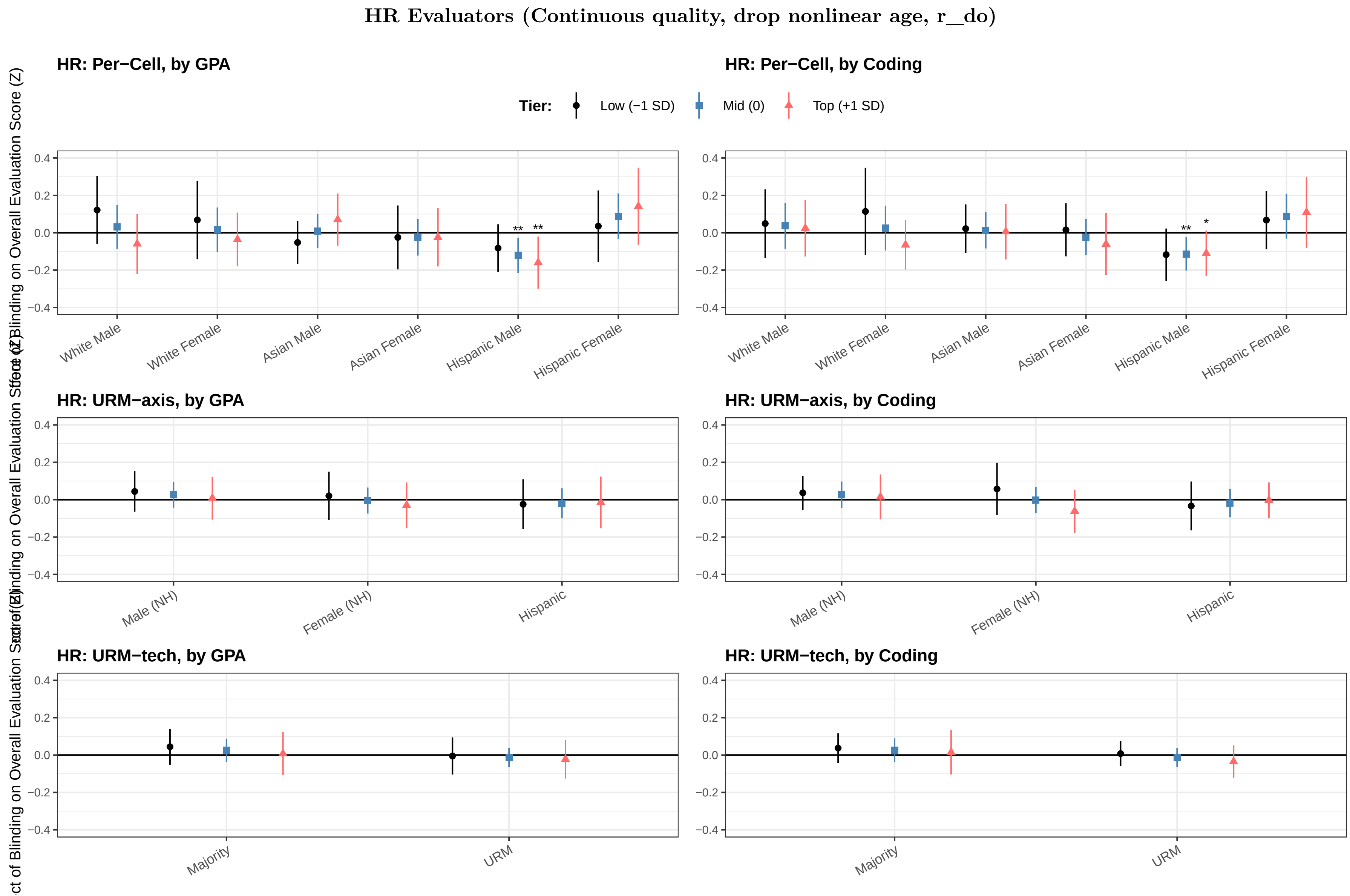
Cluster-robust standard errors (by respondent) shown in parentheses next to each $\hat{\beta}$.
Stars on p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.2 Does Blinding Lead to Selection of More Productive Coders?

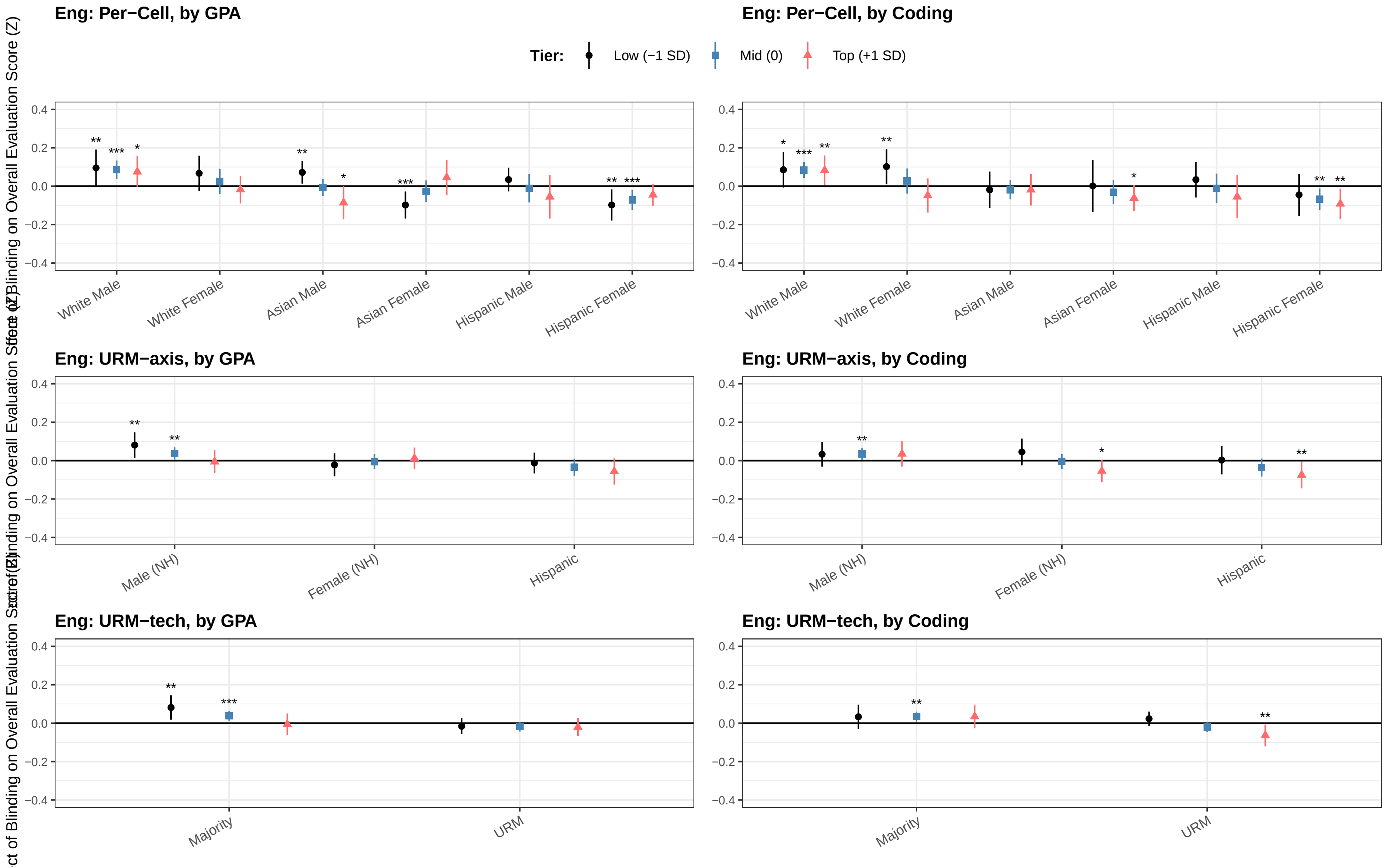


6 Heterogeneity in Treatment Effects by Candidate Quality

Spec. Single pooled regression per (demographic, quality) pair: `-sc_overall_z` $\sim$ `factor*quality_z + linear age` | `resume_index + r_do + resp FEs`, with `quality_z` $\in$ {`gpa_z`, `test_case_z`} as the standardized continuous quality variable and `factor` $\in$ {`race_gender` (6 cells), `urm_axis_blind` (Male NH / Female NH / Hispanic), `urm_tech_blind` (Majority / URM)}. Two-way cluster on `responseid + resume_index`. Plotted TE points are at -1, 0, $+1$ SD of the standardized quality score (labelled "Low / Mid / Top") via the linear combination $\hat{\beta}_D + q \cdot \hat{\beta}_{D \times \text{quality}}$, $q \in \{-1, 0, 1\}$.



Engineer Evaluators (Continuous quality, drop nonlinear age, r__do)



7 Inter-Departmental Backlash and Compensation

Spec. Two pooled regressions per role: `-sc_overall_z` $\sim$ `factor*perception + linear age` | `resume_index + r_do + resp FEs`, two-way cluster on `responseid + resume_index`. TE per (demographic, perception bucket) recovered as $\hat{\beta}_D + p \cdot \hat{\beta}_{D\times\text{perception}}$, $p \in \{0, 1\}$.

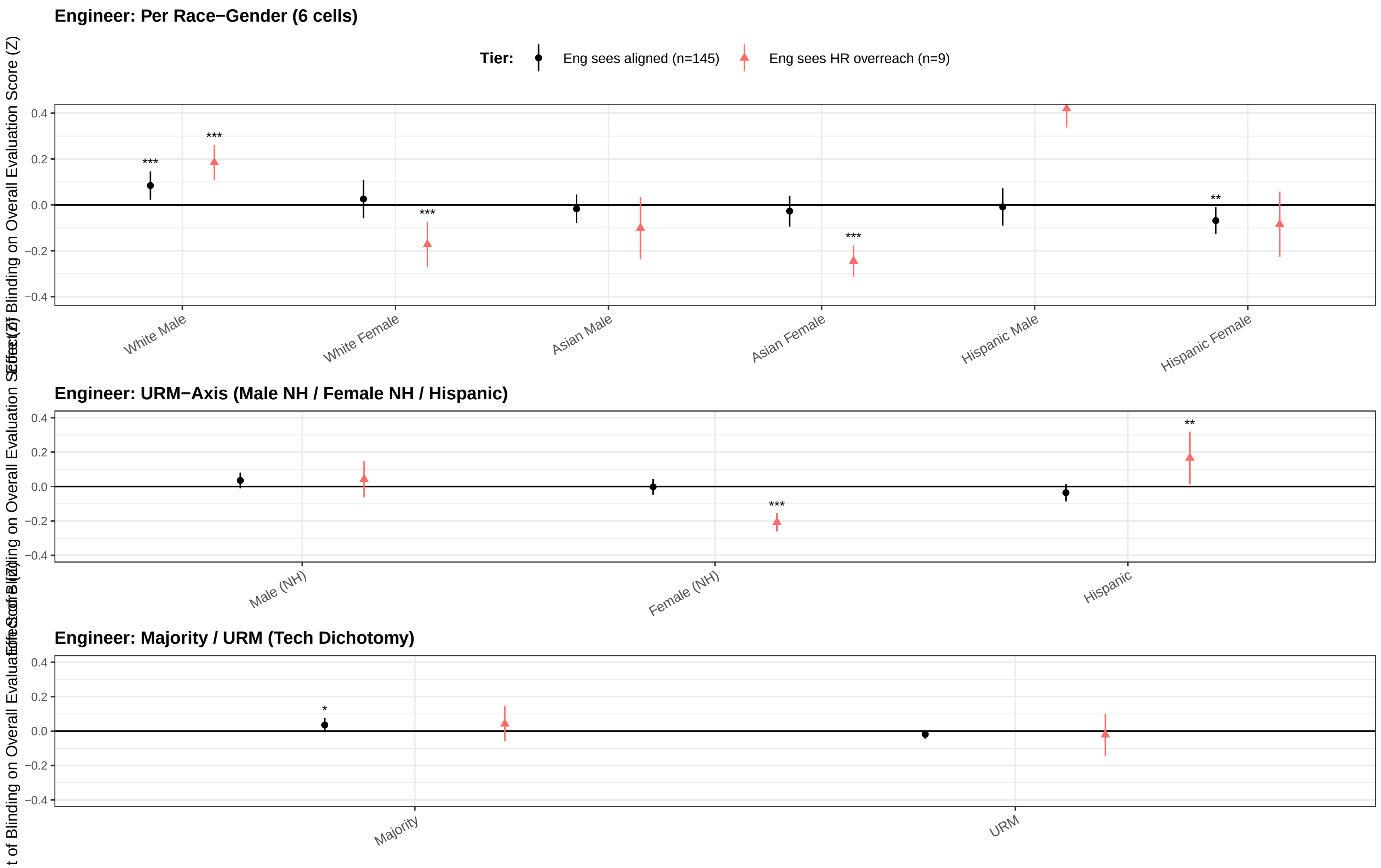
Engineer backlash (perception = `eng_perceives_hr_overreach`, =1 if Engineer marked "HR cares more about race or gender DEI than I do"; n=9 of 154 Engineers — **8 in the blind arm, only 1 in the nonblind arm**): does perceived HR-overreach trigger a counter-weight against URM candidates in nonblind?

HR compensation (perception = `hr_sees_eng_apathy`, =1 if HR rated `s5_engineer_care` as "Not at all" or "Very little"; n=33 of 137 HR — 19 blind / 14 nonblind): does perceived Engineer-apathy trigger HR over-inflation of URM scores?

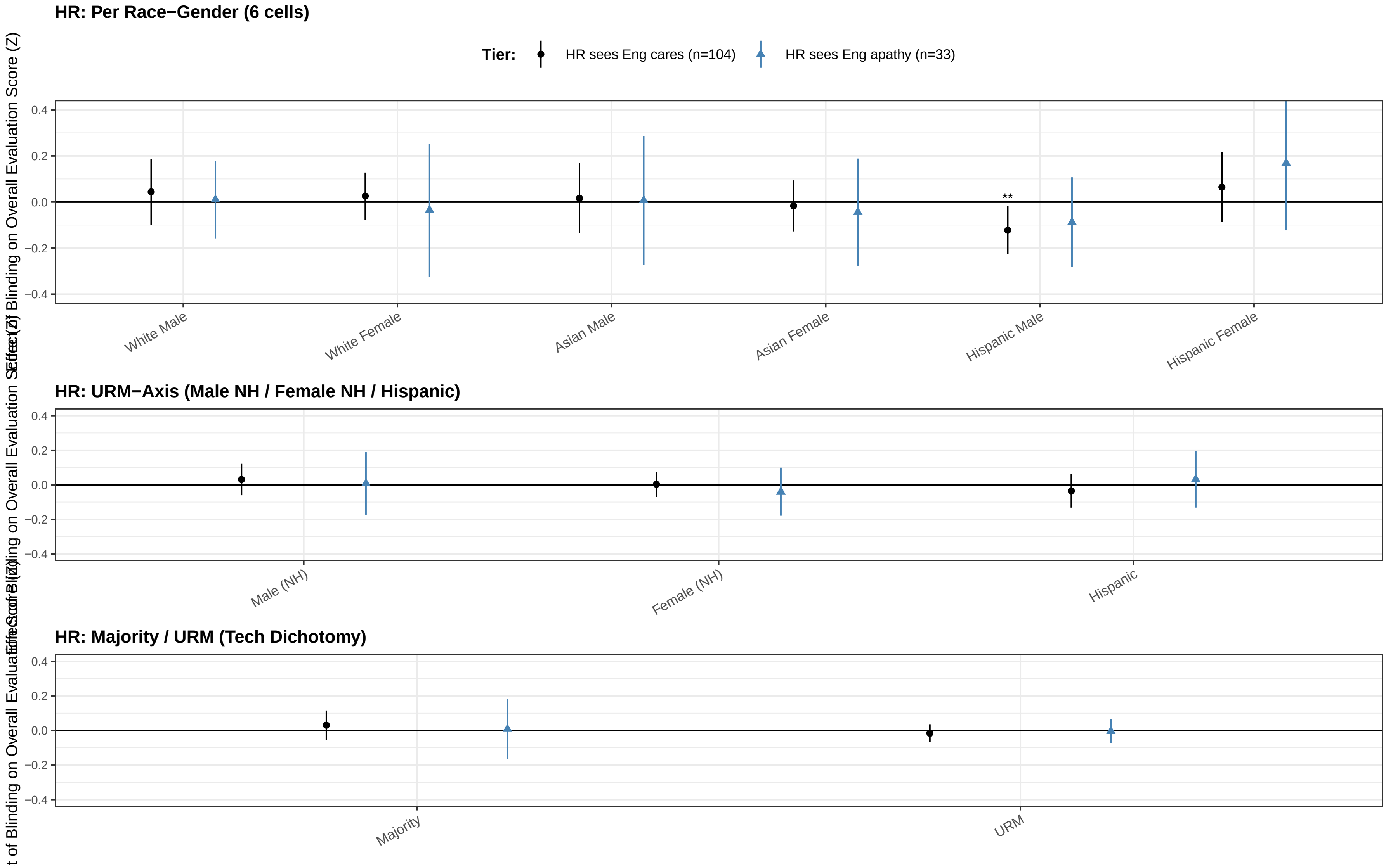
Each role's figure shows three stacked panels: per race-gender (6 cells), URM-axis trichotomy (Male NH / Female NH / Hispanic), and Majority / URM dichotomy.

Critical caveat for Engineer panels: the rebellion-bucket treatment effects are identified almost entirely from **one** respondent: only 1 of the 9 rebel Engineers was randomized into the nonblind arm. The reported magnitudes (e.g., Hispanic Male coefficient ≈ 0.42) and the joint Wald F-statistics on the Engineer side reflect this single respondent's idiosyncratic ratings; they should not be interpreted as evidence of population-level backlash. The HR panels (19 blind / 14 nonblind in the apathy bucket) are not subject to this concern.

Engineer Backlash Hypothesis


Engineer: Per Race−Gender (6 cells)
Tier:
Eng sees aligned (n=145)
Eng sees HR overreach (n=9)
Effect of Blinding on Overall Evaluation Score (Z)
0.4
0.2
0.0
−0.2
−0.4
***
***
***
***
**
White Male
White Female
Asian Male
Asian Female
Hispanic Male
Hispanic Female
Engineer: URM−Axis (Male NH / Female NH / Hispanic)
**
***
Male (NH)
Female (NH)
Hispanic
Engineer: Majority / URM (Tech Dichotomy)
*
Majority
URM

HR Compensation Hypothesis



8 Inter-Departmental Backlash and Compensation: Triple-Interaction Refinements

Spec. Three-way interaction extending §7. Each role is fit on its own sample with the role-specific perception indicator (re-named `perc_role` so HR and Eng coefs share a name): `sc_overall_z` $\sim$ `perc_role + Z + I(perc_role · Z) + i(factor, ref='Blind') + i(factor, perc_role) + i(factor, Z) + i(factor, I(perc_role · Z)) + resp_age | resume_index + r_do + resp FEs`, two-way cluster on `responseid + resume_index`. Factor $\in$ {`race_gender` (6 cells), `urm_axis_blind` (Male NH / Female NH / Hispanic), `urm_tech_blind` (Majority / URM in Tech)}. Z is alternately the respondent's own DEI Index (`dei_index_A_z`) or the candidate's coding ability (`test_case_z`). **All coefficients flipped post-fit so positive = effect of blinding raises that group's score.**

Reading the table. For each demographic D, the four blocks are:

- **D (single):** effect of blinding for D when Perc = 0 and Z = 0 (the baseline "Aligned" effect).
- **D × Perc:** how that effect shifts when Perc = 1 (Eng "rebels" / HR "compensators").
- **D × Z:** slope of the effect of blinding in Z when Perc = 0.
- **D × Perc × Z:** additional slope shift when Perc = 1. This parameter drives the scissors widening or closing.

Carry-over caveat (from §7). The Engineer rebellion bucket has only **1 of 9 rebels in the nonblind arm**. Every Eng `Misaligned'' (Perc = 1) coefficient and every Eng`Misaligned'' line in the scissors plots is identified essentially off that single respondent's ratings. Treat magnitudes as anecdotal; the 95% CI bands rely on the full cluster sandwich and *understate* true uncertainty for the rebellion line. The HR compensation bucket (19 blind / 14 nonblind) is not subject to this concern.

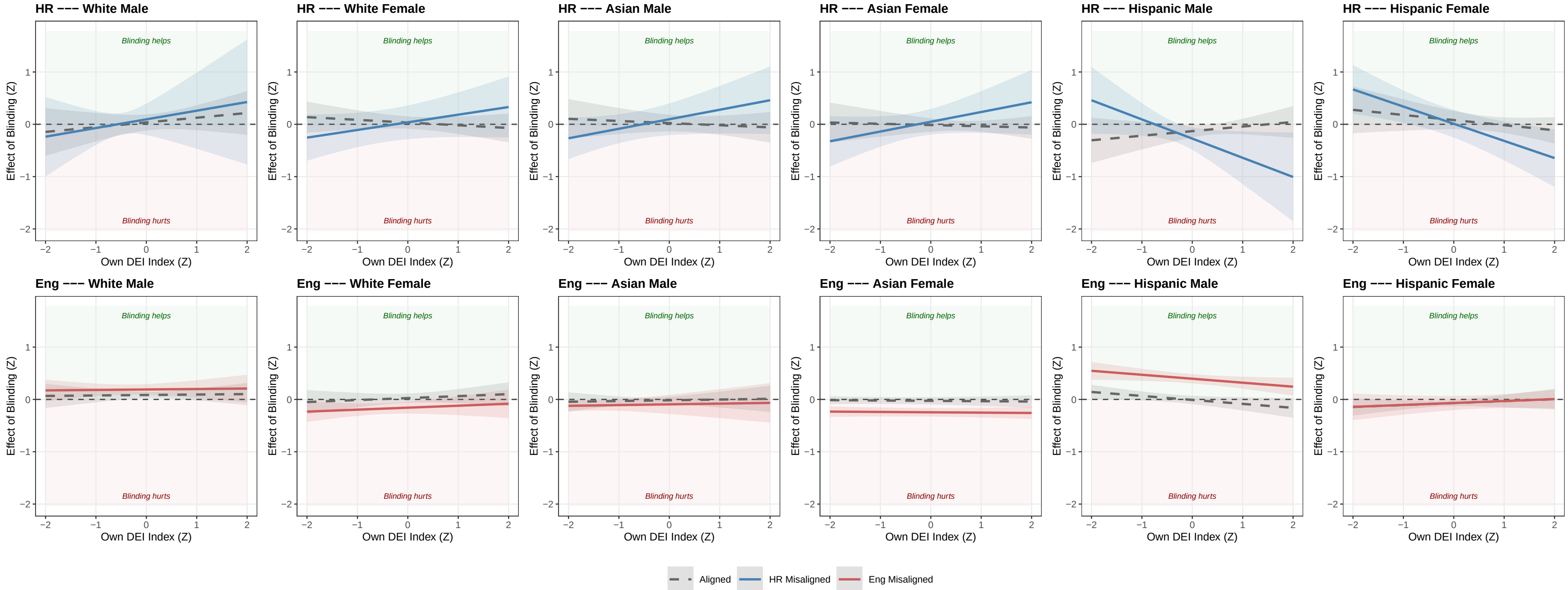
Identification note for Eng panels. With only one nonblind rebel, the variable `perc_role` · Z collapses to a constant times `perc_role` for that respondent and is collinear with the lower-order interactions, so `fixest` drops all three D × Perc × Z coefficients (see `$collin.var`). The plotted Eng `Misaligned'' line therefore inherits the slope of the`Aligned'' line and differs only by the constant Demo × Perc shift — the two lines are mechanically parallel. No *differential* Z slope is identified for Engineer; visual divergence between the lines is purely the perc-bucket intercept shift.

8.1 Per-Race-Gender (6 cells)

	Simple Perc Heterogeneity		Triple: Perc x Own DEI		Triple: Perc x Coding	
	HR	Engineer	HR	Engineer	HR	Engineer
White Male	0.044 (0.073)	0.084** (0.031)	0.035 (0.078)	0.084** (0.032)	0.044 (0.073)	0.083*** (0.027)
White Female	0.026 (0.052)	0.026 (0.043)	0.033 (0.060)	0.025 (0.045)	0.045 (0.065)	0.029 (0.038)
Asian Male	0.016 (0.077)	−0.017 (0.032)	0.023 (0.082)	−0.016 (0.035)	0.008 (0.077)	−0.018 (0.030)
Asian Female	−0.017 (0.056)	−0.027 (0.034)	−0.014 (0.065)	−0.027 (0.037)	−0.016 (0.065)	−0.027 (0.037)
Hispanic Male	−0.123** (0.053)	−0.009 (0.042)	−0.131** (0.061)	−0.009 (0.041)	−0.123** (0.055)	−0.015 (0.043)
Hispanic Female	0.064 (0.077)	−0.068** (0.029)	0.081 (0.090)	−0.069** (0.030)	0.066 (0.076)	−0.068* (0.034)
White Male × Misaligned (Perc=1)	−0.034 (0.099)	0.101*** (0.028)	0.059 (0.154)	0.106** (0.043)	−0.035 (0.118)	0.269*** (0.054)
White Female × Misaligned (Perc=1)	−0.061 (0.143)	−0.198*** (0.033)	0.004 (0.162)	−0.183*** (0.038)	−0.086 (0.153)	−0.154*** (0.045)
Asian Male × Misaligned (Perc=1)	−0.009 (0.195)	−0.084 (0.062)	0.073 (0.205)	−0.078 (0.081)	0.005 (0.148)	−0.006 (0.082)
Asian Female × Misaligned (Perc=1)	−0.027 (0.127)	−0.218*** (0.041)	0.062 (0.130)	−0.219*** (0.041)	−0.025 (0.131)	−0.214*** (0.052)
Hispanic Male × Misaligned (Perc=1)	0.035 (0.113)	0.429*** (0.025)	−0.143 (0.106)	0.405*** (0.032)	0.038 (0.117)	0.525*** (0.037)
Hispanic Female × Misaligned (Perc=1)	0.105 (0.160)	−0.016 (0.085)	−0.070 (0.167)	0.001 (0.078)	0.080 (0.149)	1.777*** (0.379)
White Male × Own DEI Index (Z)			0.090 (0.105)	0.009 (0.054)		
White Female × Own DEI Index (Z)			−0.052 (0.066)	0.038 (0.054)		
Asian Male × Own DEI Index (Z)			−0.041 (0.076)	0.014 (0.053)		
Asian Female × Own DEI Index (Z)			−0.024 (0.072)	−0.007 (0.017)		
Hispanic Male × Own DEI Index (Z)			0.088 (0.090)	−0.076* (0.037)		
Hispanic Female × Own DEI Index (Z)			−0.098 (0.081)	0.037 (0.044)		
White Male × Candidate Coding (Z)					0.001 (0.064)	0.009 (0.039)
White Female × Candidate Coding (Z)					−0.090 (0.092)	−0.062* (0.034)
Asian Male × Candidate Coding (Z)					0.084 (0.076)	0.009 (0.039)
Asian Female × Candidate Coding (Z)					−0.017 (0.068)	−0.021 (0.045)
Hispanic Male × Candidate Coding (Z)					0.001 (0.070)	−0.032 (0.035)
Hispanic Female × Candidate Coding (Z)					0.082 (0.073)	−0.012 (0.041)
White Male × Misaligned (Perc=1) × Own DEI Index (Z)			0.075 (0.244)			
White Female × Misaligned (Perc=1) × Own DEI Index (Z)			0.198 (0.121)			
Asian Male × Misaligned (Perc=1) × Own DEI Index (Z)			0.223 (0.133)			
Asian Female × Misaligned (Perc=1) × Own DEI Index (Z)			0.210 (0.146)			
Hispanic Male × Misaligned (Perc=1) × Own DEI Index (Z)			−0.454** (0.196)			
Hispanic Female × Misaligned (Perc=1) × Own DEI Index (Z)			−0.230* (0.131)			
White Male × Misaligned (Perc=1) × Candidate Coding (Z)					−0.039 (0.127)	−0.378*** (0.061)
White Female × Misaligned (Perc=1) × Candidate Coding (Z)					−0.003 (0.138)	−0.265*** (0.065)
Asian Male × Misaligned (Perc=1) × Candidate Coding (Z)					−0.383** (0.150)	−0.040 (0.112)
Asian Female × Misaligned (Perc=1) × Candidate Coding (Z)					−0.091 (0.120)	−0.114* (0.058)
Hispanic Male × Misaligned (Perc=1) × Candidate Coding (Z)					0.022 (0.123)	0.027 (0.070)
Hispanic Female × Misaligned (Perc=1) × Candidate Coding (Z)					−0.267 (0.163)	−2.011*** (0.426)
Resume (Content) FE	✓	✓	✓	✓	✓	✓
Display Position FE	✓	✓	✓	✓	✓	✓
Respondent FE	×	×	×	×	×	×
Respondent Characteristics	✓	✓	✓	✓	✓	✓
Observations	2466	2772	2466	2772	2466	2772

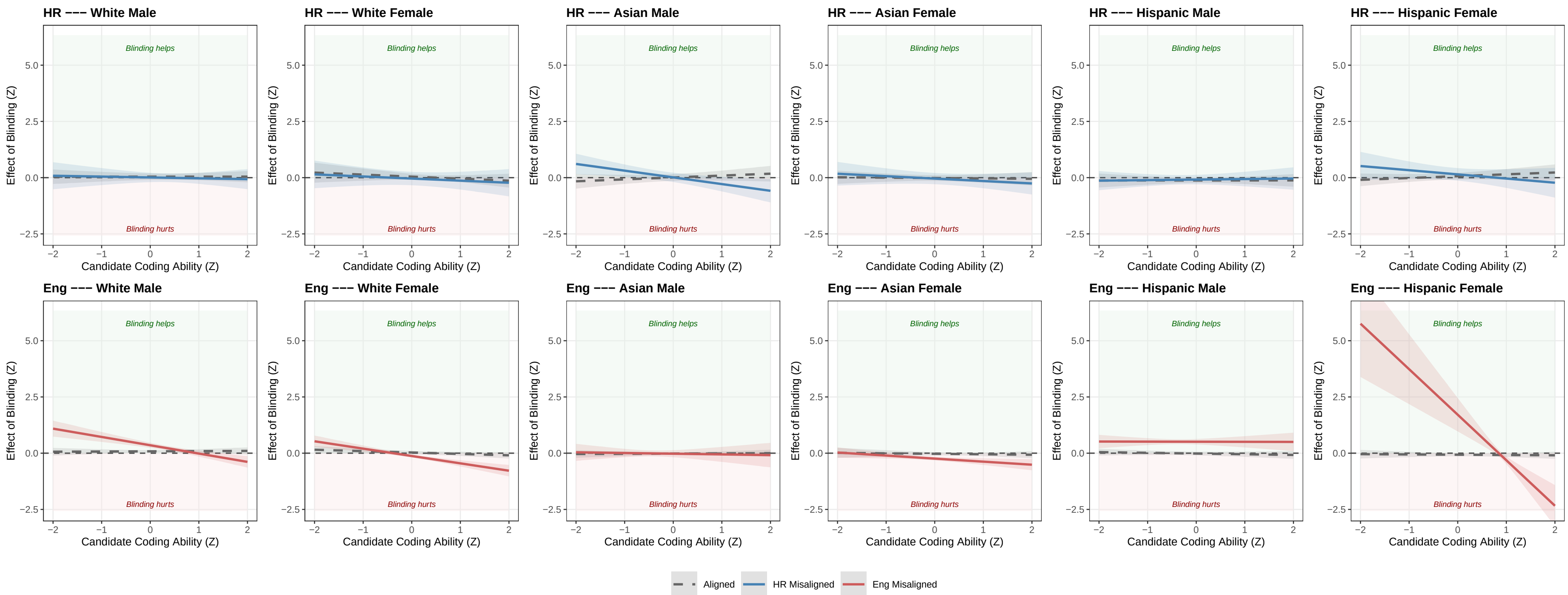
HR Perc=1 means HR sees Eng apathy (n=33 apathy / 104 cares). Eng Perc=1 means Eng perceives HR overreach (n=9 rebels / 145 aligned; only 1 rebel in nonblind arm). Coefficients flipped post-fit so positive = effect of blinding raises the score. Two-way cluster on responseid and resume index.

Triple A: Perc × Own DEI Index — Per Race-Gender (6 cells)



Top row: HR (Aligned = HR sees Eng cares; Misaligned = HR sees Eng apathy). Bottom row: Engineer (Aligned = view-aligned with HR; Misaligned = perceives HR overreach). Lines: predicted effect of blinding on each demographic's overall score as a function of the rater's own DEI Index (Z); shaded bands = 95% CI from the linear combination of triple-interaction coefs (SE clustered two-way on responseid + resume_index). **Eng "Misaligned" line is identified essentially off 1 nonblind respondent — treat as anecdotal.**

Triple B: Perc × Candidate Coding Ability — Per Race-Gender (6 cells)



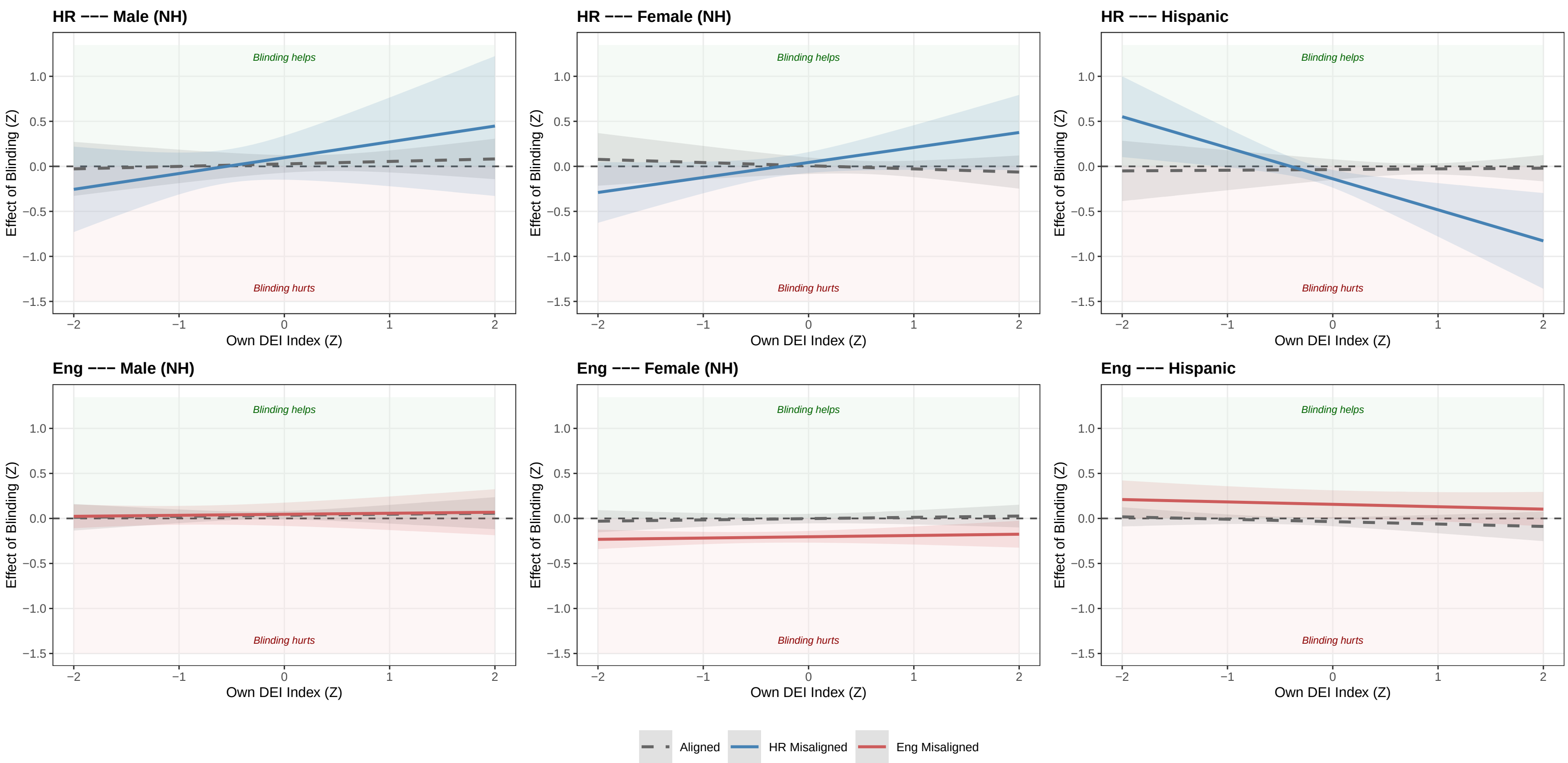
Same panel layout as Triple A. X-axis: candidate's coding ability (test_case z-score). Same Eng-rebellion caveat applies.

8.2 URM-Axis Trichotomy (Male NH / Female NH / Hispanic)

	Simple Perc Heterogeneity		Triple: Perc x Own DEI		Triple: Perc x Coding	
	HR	Engineer	HR	Engineer	HR	Engineer
Male (NH)	0.031 (0.047)	0.035 (0.024)	0.027 (0.050)	0.035 (0.024)	0.031 (0.046)	0.034 (0.022)
Female (NH)	0.003 (0.037)	−0.002 (0.023)	0.007 (0.047)	−0.002 (0.027)	0.009 (0.046)	−0.002 (0.023)
Hispanic	−0.035 (0.049)	−0.036 (0.026)	−0.035 (0.059)	−0.035 (0.026)	−0.031 (0.042)	−0.039 (0.027)
Male (NH) × Misaligned (Perc=1)	−0.023 (0.113)	0.007 (0.032)	0.069 (0.139)	0.011 (0.047)	−0.009 (0.104)	0.027 (0.030)
Female (NH) × Misaligned (Perc=1)	−0.042 (0.078)	−0.206*** (0.010)	0.036 (0.072)	−0.202*** (0.016)	−0.050 (0.088)	−0.188*** (0.027)
Hispanic × Misaligned (Perc=1)	0.067 (0.085)	0.202** (0.082)	−0.103 (0.065)	0.192** (0.081)	0.060 (0.086)	0.216*** (0.065)
Male (NH) × Own DEI Index (Z)			0.028 (0.063)	0.011 (0.040)		
Female (NH) × Own DEI Index (Z)			−0.035 (0.058)	0.014 (0.029)		
Hispanic × Own DEI Index (Z)			0.008 (0.059)	−0.027 (0.033)		
Male (NH) × Candidate Coding (Z)					0.037 (0.050)	0.010 (0.033)
Female (NH) × Candidate Coding (Z)					−0.046 (0.070)	−0.038 (0.029)
Hispanic × Candidate Coding (Z)					0.044 (0.051)	−0.026 (0.028)
Male (NH) × Misaligned (Perc=1) × Own DEI Index (Z)			0.148 (0.148)			
Female (NH) × Misaligned (Perc=1) × Own DEI Index (Z)			0.202* (0.112)			
Hispanic × Misaligned (Perc=1) × Own DEI Index (Z)			−0.352*** (0.121)			
Male (NH) × Misaligned (Perc=1) × Candidate Coding (Z)					−0.205* (0.102)	−0.018 (0.065)
Female (NH) × Misaligned (Perc=1) × Candidate Coding (Z)					−0.056 (0.085)	−0.186*** (0.048)
Hispanic × Misaligned (Perc=1) × Candidate Coding (Z)					−0.122 (0.101)	−0.322*** (0.087)
Resume (Content) FE	✓	✓	✓	✓	✓	✓
Display Position FE	✓	✓	✓	✓	✓	✓
Respondent FE	×	×	×	×	×	×
Respondent Characteristics	✓	✓	✓	✓	✓	✓
Observations	2466	2772	2466	2772	2466	2772

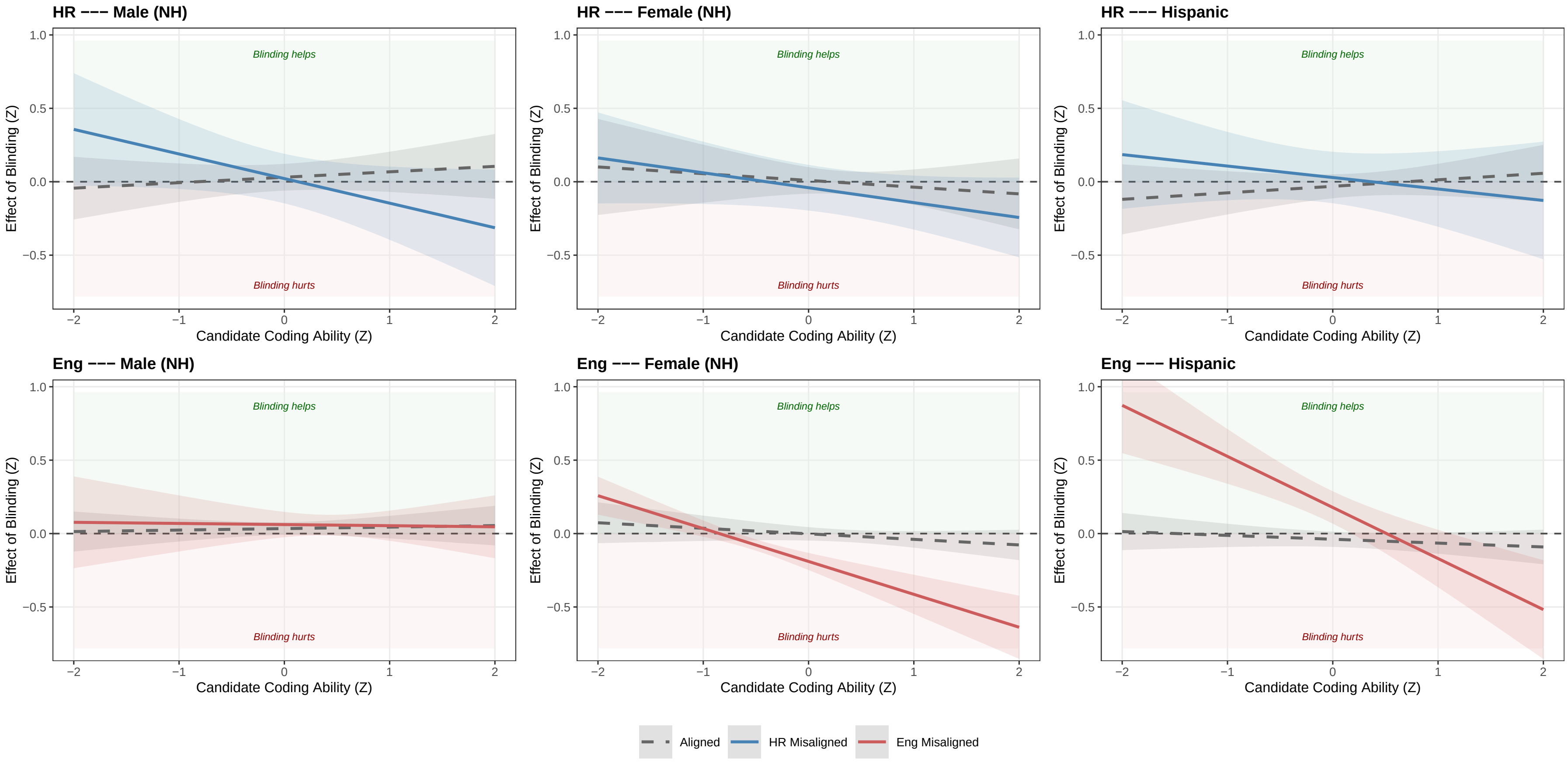
HR Perc=1 means HR sees Eng apathy (n=33 apathy / 104 cares). Eng Perc=1 means Eng perceives HR overreach (n=9 rebels / 145 aligned; only 1 rebel in nonblind arm). Coefficients flipped post-fit so positive = effect of blinding raises the score. Two-way cluster on responseid and resume index.

Triple A: Perc × Own DEI Index — URM-Axis Trichotomy



Top row: HR. Bottom row: Engineer. "Aligned" = Perc=0; "Misaligned" = Perc=1 (HR-sees-Eng-apathy / Eng-sees-HR-overreach). Bands = 95% CI. **Eng "Misaligned" identified off 1 nonblind respondent — treat as anecdotal.**

Triple B: Perc × Candidate Coding Ability — URM-Axis Trichotomy



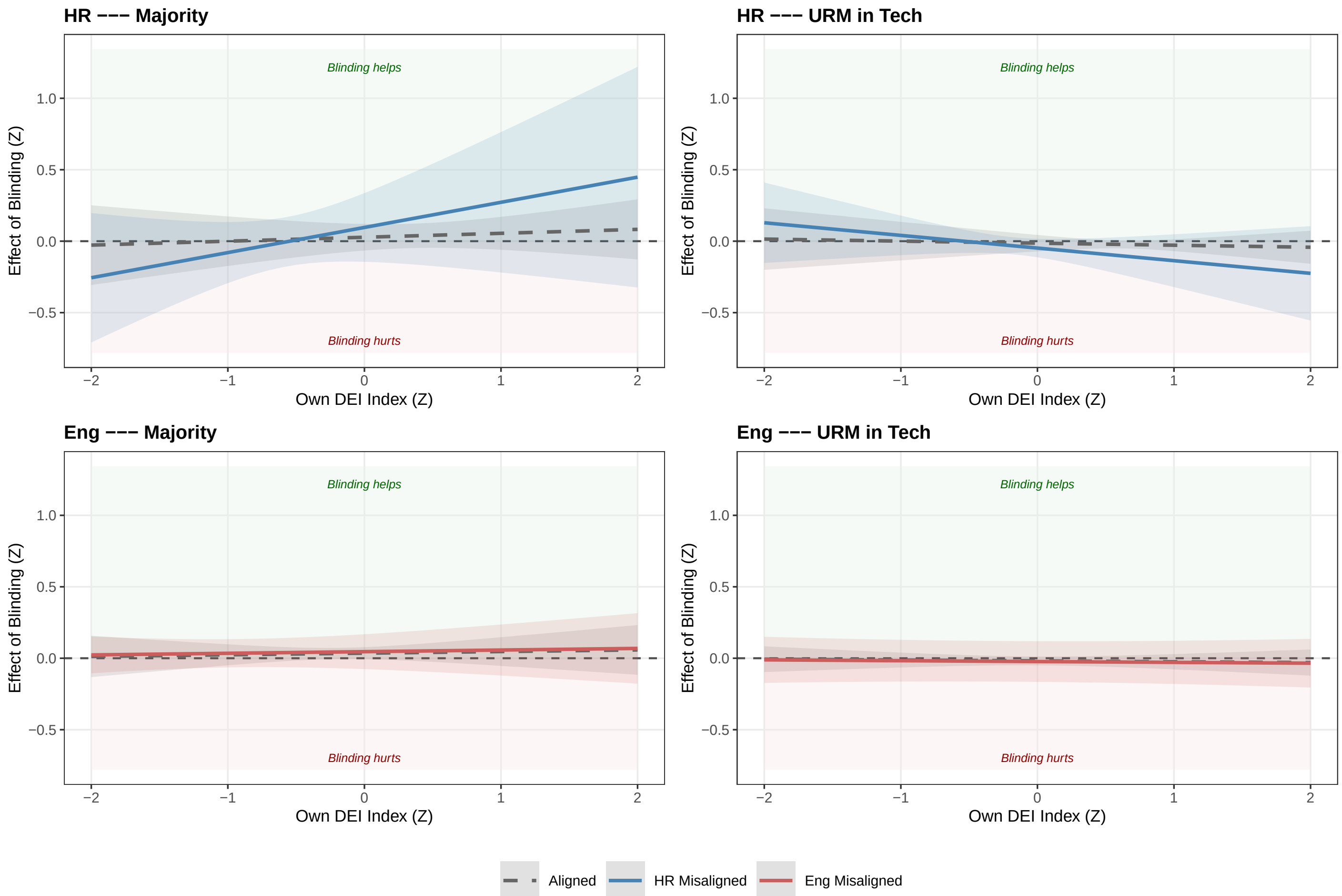
Same layout as Triple A. X-axis: candidate's coding ability (test_case z-score). Same Eng-rebellion caveat applies.

8.3 URM-Tech Dichotomy (Majority in Tech / URM in Tech)

	Simple Perc Heterogeneity		Triple: Perc x Own DEI		Triple: Perc x Coding	
	HR	Engineer	HR	Engineer	HR	Engineer
Majority	0.031 (0.043)	0.035 (0.021)	0.027 (0.047)	0.035 (0.021)	0.031 (0.042)	0.034* (0.018)
URM in Tech	−0.016 (0.025)	−0.019 (0.013)	−0.014 (0.029)	−0.019 (0.016)	−0.016 (0.027)	−0.020 (0.016)
Majority × Misaligned (Perc=1)	−0.022 (0.112)	0.007 (0.032)	0.069 (0.140)	0.011 (0.044)	−0.010 (0.099)	0.027 (0.028)
URM in Tech × Misaligned (Perc=1)	0.012 (0.030)	−0.002 (0.055)	−0.034 (0.049)	−0.004 (0.064)	0.009 (0.043)	0.016 (0.048)
Majority × Own DEI Index (Z)			0.028 (0.058)	0.011 (0.039)		
URM in Tech × Own DEI Index (Z)			−0.014 (0.042)	−0.006 (0.022)		
Majority × Candidate Coding (Z)					0.036 (0.048)	0.010 (0.030)
URM in Tech × Candidate Coding (Z)					0.001 (0.035)	−0.031 (0.022)
Majority × Misaligned (Perc=1) × Own DEI Index (Z)			0.148 (0.142)			
URM in Tech × Misaligned (Perc=1) × Own DEI Index (Z)			−0.074 (0.071)			
Majority × Misaligned (Perc=1) × Candidate Coding (Z)					−0.203* (0.098)	−0.018 (0.063)
URM in Tech × Misaligned (Perc=1) × Candidate Coding (Z)					−0.093* (0.052)	−0.255*** (0.047)
Resume (Content) FE	✓	✓	✓	✓	✓	✓
Display Position FE	✓	✓	✓	✓	✓	✓
Respondent FE	×	×	×	×	×	×
Respondent Characteristics	✓	✓	✓	✓	✓	✓
Observations	2466	2772	2466	2772	2466	2772

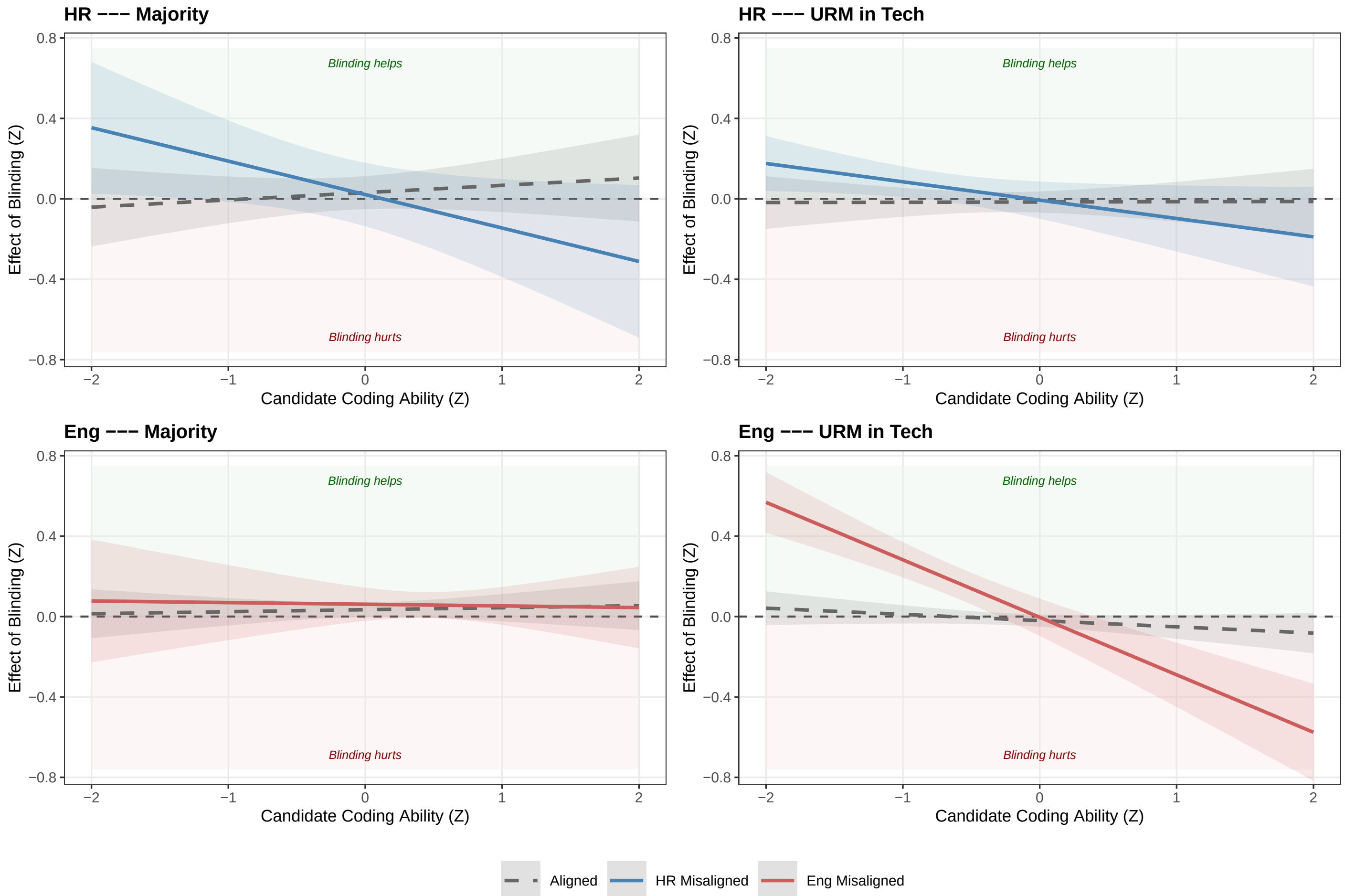
HR Perc=1 means HR sees Eng apathy (n=33 apathy / 104 cares). Eng Perc=1 means Eng perceives HR overreach (n=9 rebels / 145 aligned; only 1 rebel in nonblind arm). Coefficients flipped post-fit so positive = effect of blinding raises the score. Two-way cluster on responseid and resume index.

Triple A: Perc × Own DEI Index — URM-Tech Dichotomy



Top row: HR. Bottom row: Engineer. "Aligned" = Perc=0; "Misaligned" = Perc=1. Bands = 95% CI. **Eng "Misaligned" identified off 1 nonblind respondent — treat as anecdotal.**

Triple B: Perc × Candidate Coding Ability — URM-Tech Dichotomy



Same layout as Triple A. X-axis: candidate's coding ability (test_case z-score). Same Eng-rebellion caveat applies.